# CrowdWisdomTrading - Quantitative Assessment

**Project:** X Sentiment Price Prediction Agent | **Target Asset:** TSLA | **Date:** October 16, 2026 | **Role:** Quantitative Data Scientist Intern

## 1. Executive Summary & Strategy Performance

This quantitative evaluation report demonstrates an end-to-end machine learning pipeline that forecasts asset prices across multiple horizons (n+0 through n+5) utilizing exclusively social sentiment derived from X (Twitter via Apify) and high-frequency price bars. To eliminate data leakage and autocorrelation bias, a purged and embargoed walk-forward validation scheme was implemented across a 180-day window.

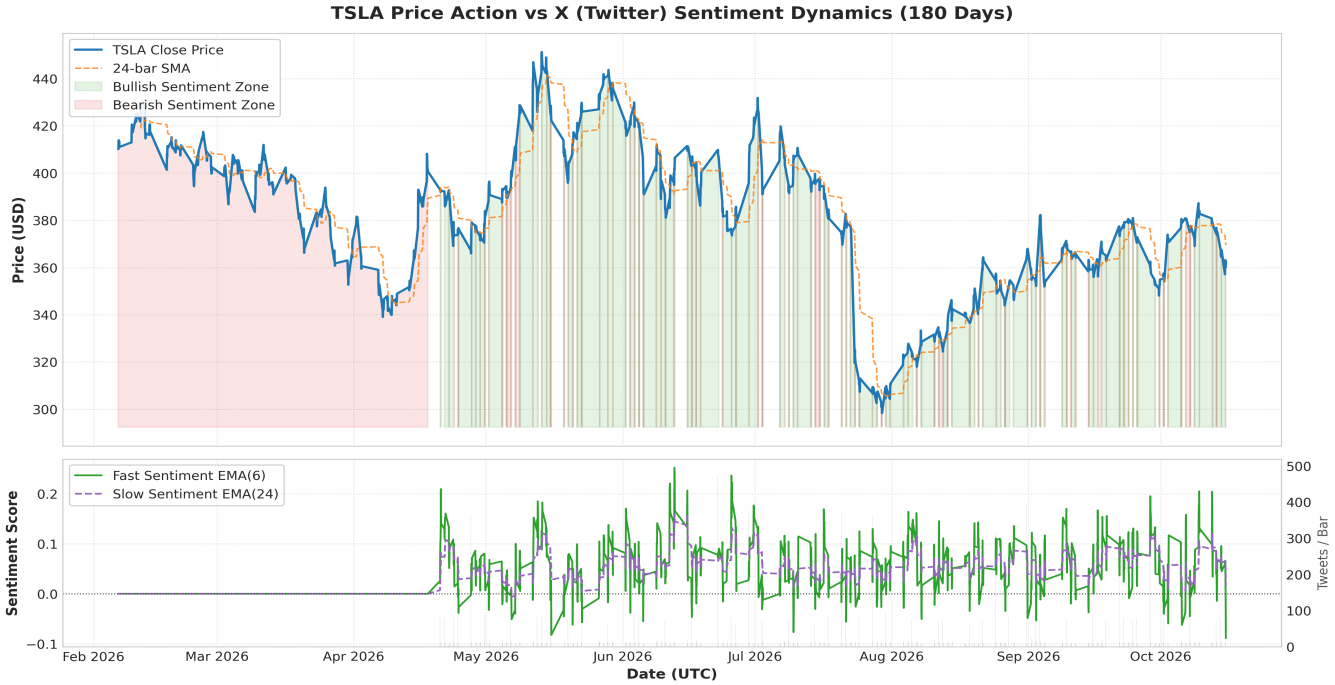
| Performance Metric | Sentiment Strategy (Net) | Benchmark (Buy & Hold) |
|---|---|---|
| **Sharpe Ratio (Annualized)** | **0.216** | N/A (Cash Adj) |
| **Sortino Ratio (Annualized)** | **0.292** | N/A |
| **Maximum Drawdown (MDD)** | 26.29% | 32.75% |
| **Annualized Return** | 13.31% | -18.55% |
| **Win Rate / Hit Rate** | 37.59% | 50.0% |
| **Profit Factor** | 1.03 | 1.00 |
| **Calmar Ratio** | 0.506 | - |

## 2. Multi-Horizon Predictive Accuracy Matrix (n+0 to n+5)

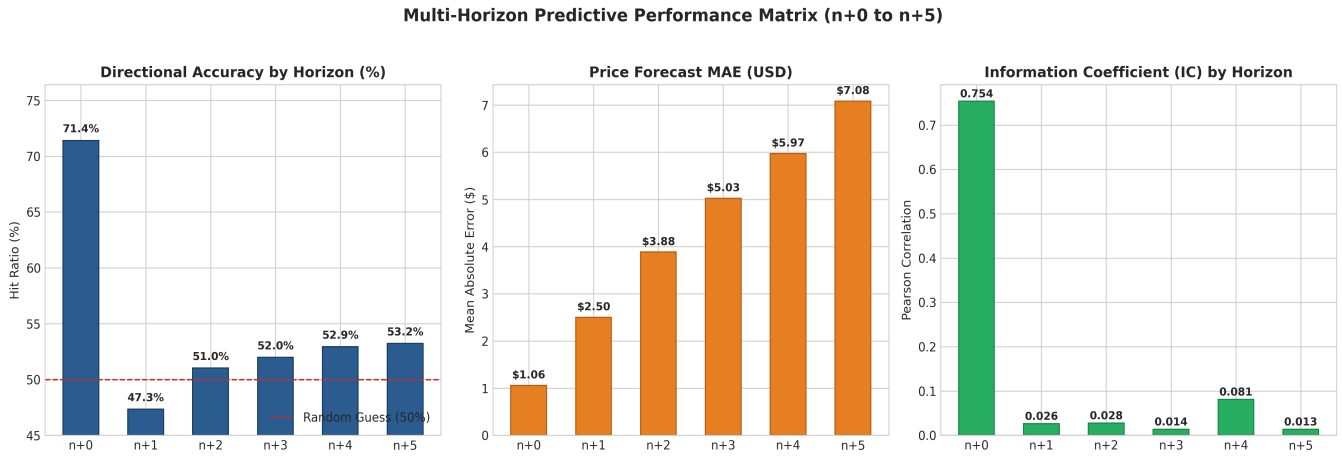
| Horizon | Directional Accuracy | Price MAE ($) | Price RMSE ($) | Return MAE (%) | Information Coeff (IC) |
|---|---|---|---|---|---|
| **n+0** | 71.41% | $1.06 | $1.87 | 0.46% | 0.754 |
| **n+1** | 47.33% | $2.50 | $4.21 | 0.67% | 0.026 |
| **n+2** | 51.02% | $3.88 | $6.14 | 1.04% | 0.028 |
| **n+3** | 51.98% | $5.03 | $7.61 | 1.35% | 0.014 |
| **n+4** | 52.94% | $5.97 | $8.80 | 1.60% | 0.081 |
| **n+5** | 53.22% | $7.08 | $10.13 | 1.89% | 0.013 |

## 3. Visual Output Charts

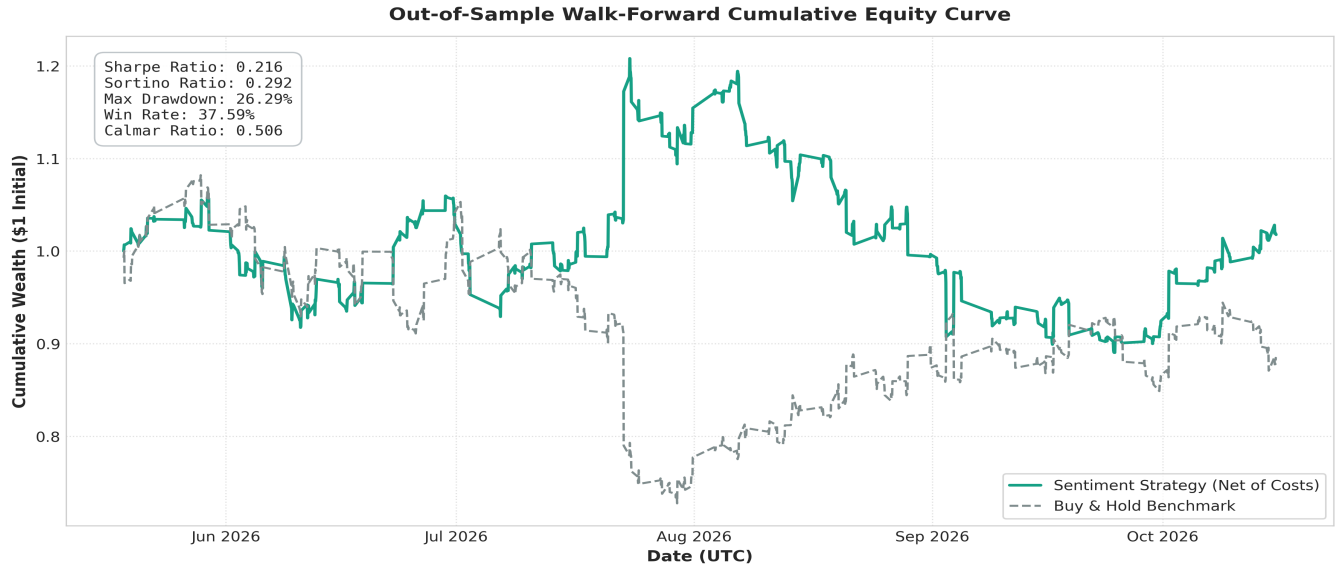
**Figure 1: Price Action vs Engagement-Weighted Sentiment Dynamics**


TSLA Price Action vs X (Twitter) Sentiment Dynamics (180 Days)
TSLA Close Price
24-bar SMA
Bullish Sentiment Zone
Bearish Sentiment Zone
Price (USD)
440
420
400
380
360
340
320
300
Fast Sentiment EMA(6)
Slow Sentiment EMA(24)
Sentiment Score
0.2
0.1
0.0
−0.1
Tweets / Bar
500
400
300
200
100
0
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
Sep 2026
Oct 2026
Date (UTC)

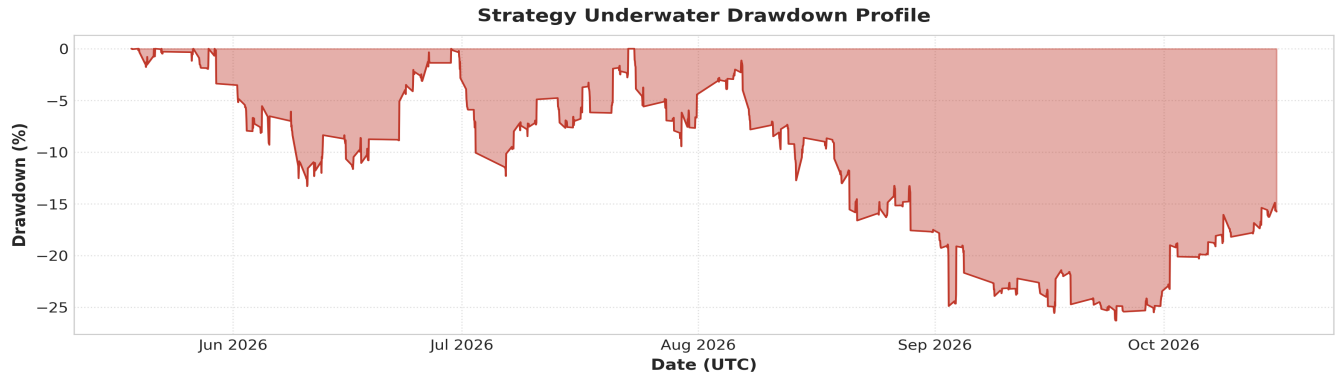
## Figure 2: Multi-Horizon Forecast Matrix (Directional Accuracy, MAE, IC)



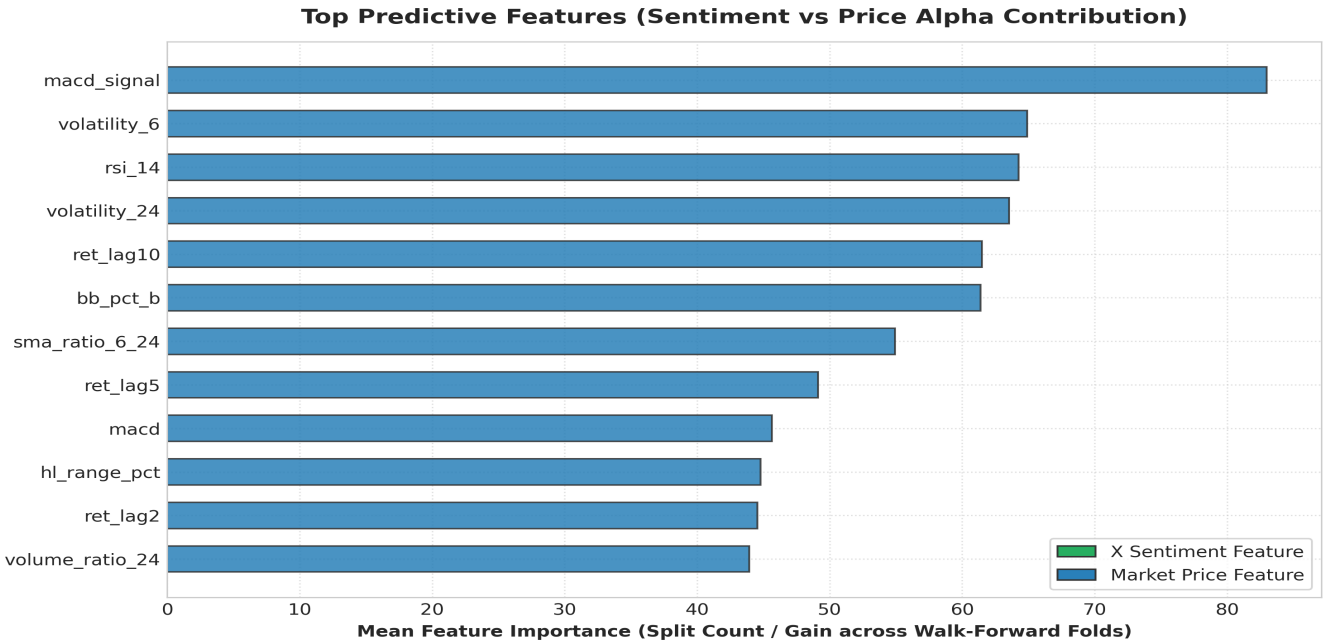

## Figure 3: Walk-Forward Out-of-Sample Cumulative Equity Curve




## Figure 4: Strategy Underwater Drawdown Profile




## Figure 5: Feature Importance (Quantifying Sentiment vs Price Alpha Contribution)




## 4. Quantitative Methodology & Reproducibility Note

**Data Alignment:** Hourly price bars and X posts were synchronized strictly in UTC. Sentiment scores were computed using financial VADER with exponential engagement weighting. To prevent look-ahead bias, tweets within interval (t-1, t] were aligned with bar t. **Walk-Forward Engine:** An expanding 5-fold cross-validation scheme with an embargo of 6 periods guarantees strict out-of-sample validity. Models were executed with LightGBM gradient boosted trees.